

Observing the Unobservable

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Supervised By

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Motivation

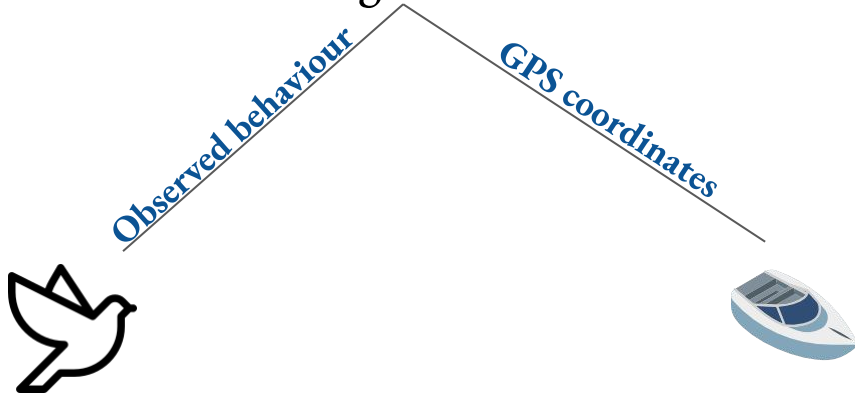
- ❑ The understanding and accurate identification of animal behaviours are helpful for conservation actions
- ❑ It is difficult to directly observe behaviours influencing animal movement in remote and hostile terrain such as the marine environment

Goal

- ❑ Better behavioural inference of seabirds from tracking data for effective conservation actions and management - identifying important foraging areas
 - validating behaviours inferred from commonly used HMMs
 - examining the effect of temporal resolution on behavioural inference

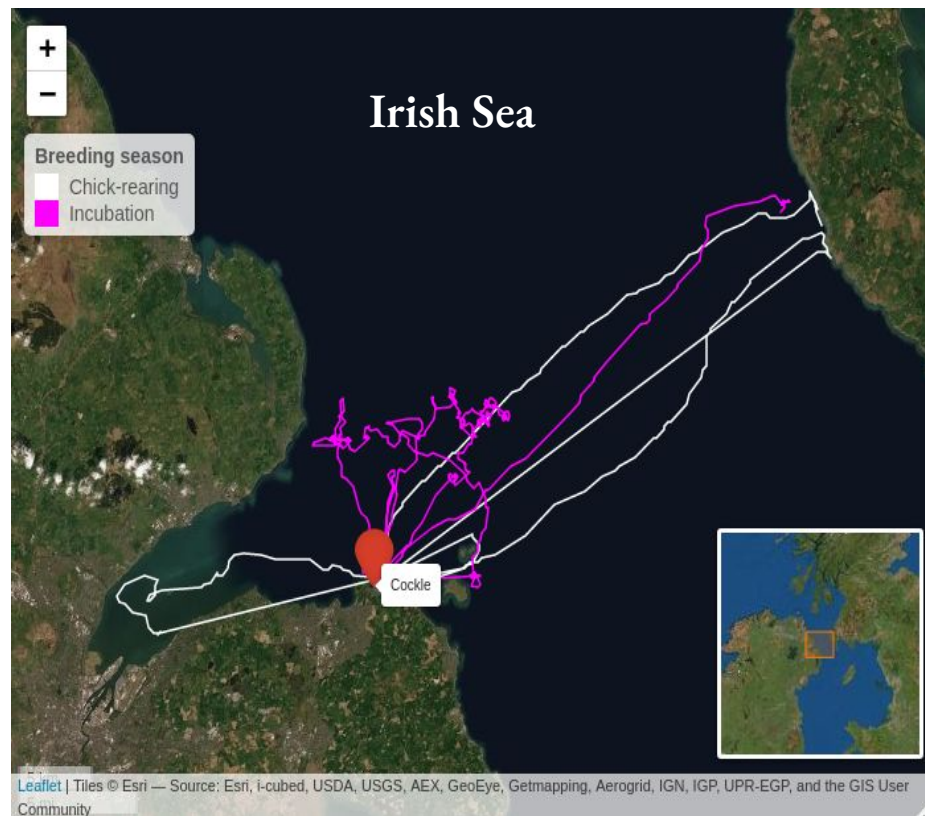
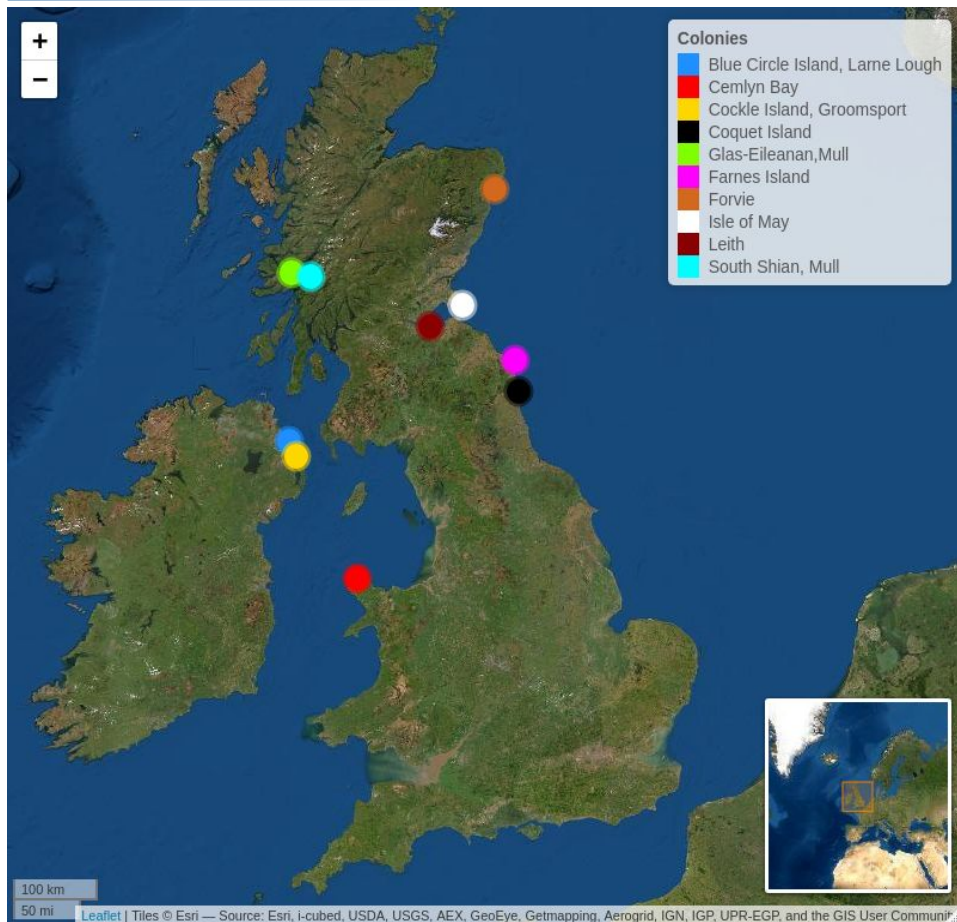
Data

- ❑ Collected in the UK by the Joint Nature Conservation Committee
- ❑ Visual tracking data of terns



- ❑ Period - breeding seasons (2009 - 2011)

Study sites



Visual tracking (boat movement) data of sandwich terns at Cockle Island colony (54°67'N, 5°61'W), Northern Ireland, 2010

Work flow

- 💡 **Assess** boat movement data as proxy to tern's track
- 💡 **Infer** hidden behaviours using HMMs
- 💡 **Validate** inferred behaviours using observed behavioural data

Method

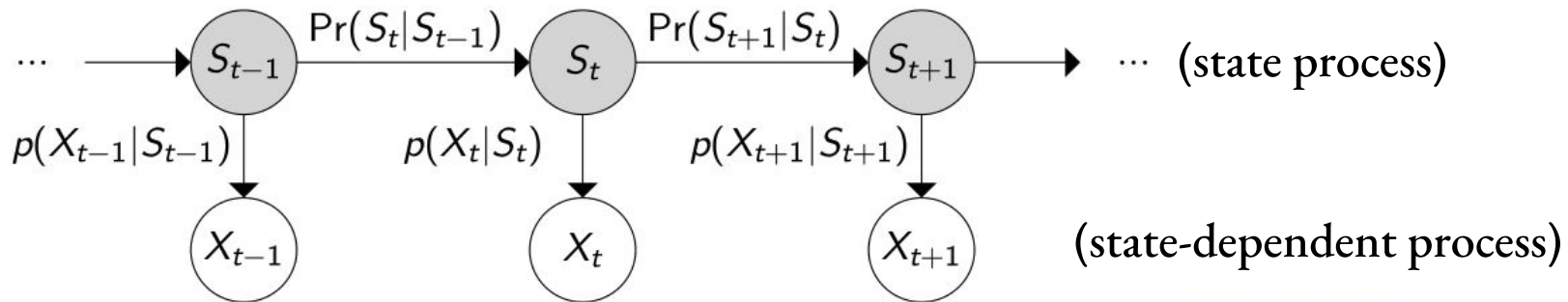


Figure: Representation of a HMM

Assessing boat movement data

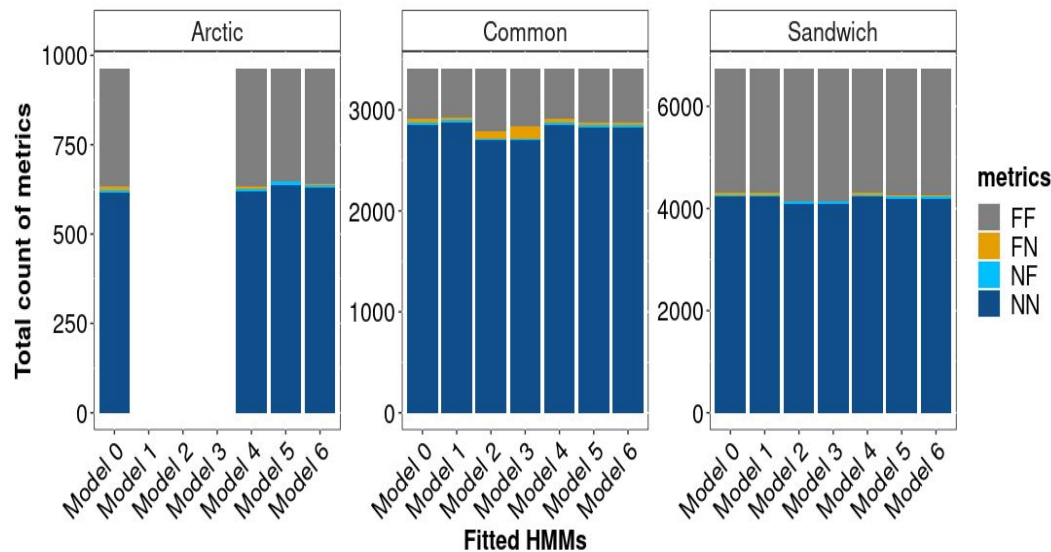


Figure 2: Confusion matrix metrics of inferred behavioural states from HMMs fitted to boat location and approximate location data of 9 terns: 1 Arctic, 2 common and 6 Sandwich, in Coquet colony during chick-rearing, 2009.

FF = Foraging/Foraging (true positive), FN = Foraging/Non-foraging (false negative), NF = Non-foraging/Foraging (false positive), NN = Non-Foraging/Non-foraging (true negative)

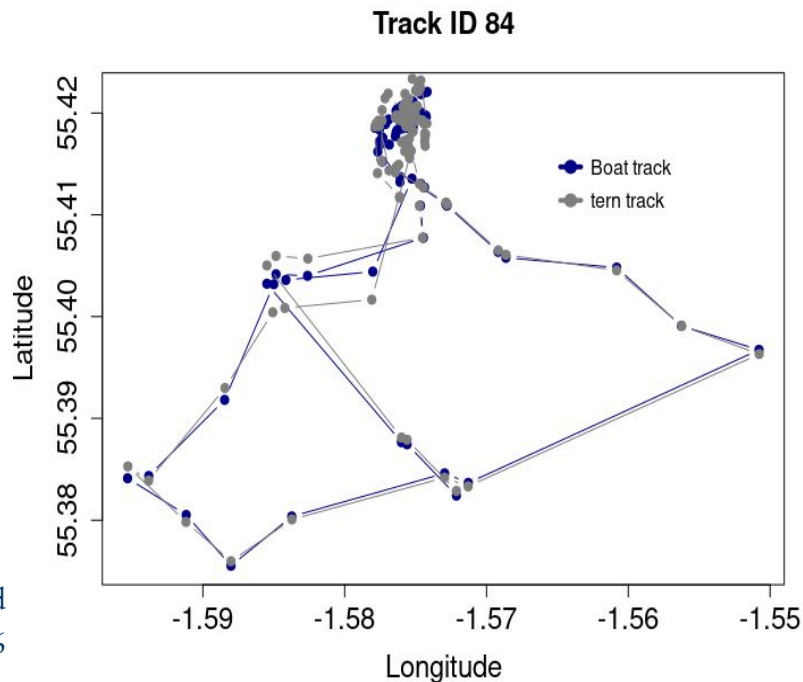


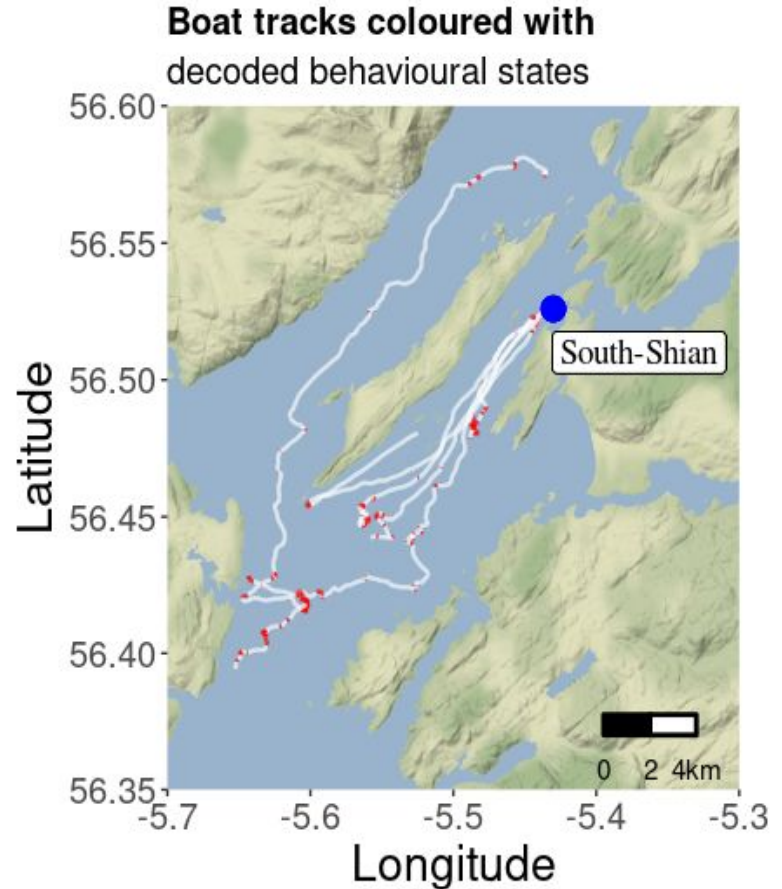
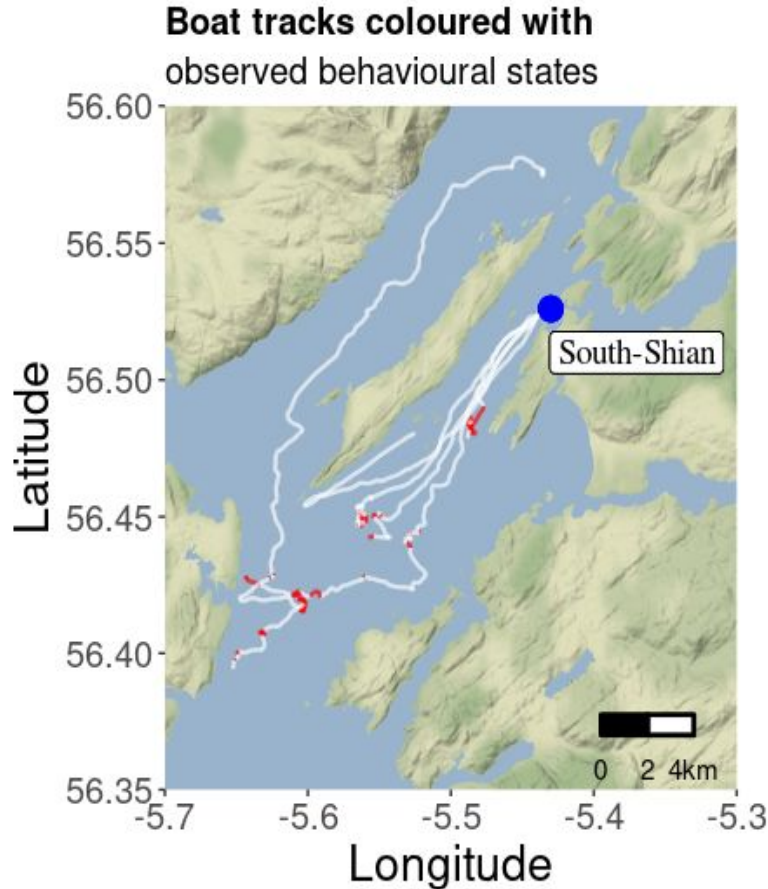
Figure 3: boat track and approximate tern track from Coquet colony

Validating inferred states

Table: Summarised results from 2-state HMMs fitted to visual tracking data of terns across study sites during chick-rearing

	HMMs deemed optimal based on least log-loss value		
	Complete pooling	Covariate effect	No pooling
F1-score	(78.48 - 83.55)%	(73.24 - 79.31)%	(71.77 - 87.86)%
Tern species	common	Arctic, common, roseate	Arctic, common, Sandwich
No. of colonies	2	6	3

Boat tracks showing observed and inferred states



Foraging events - chick rearing season

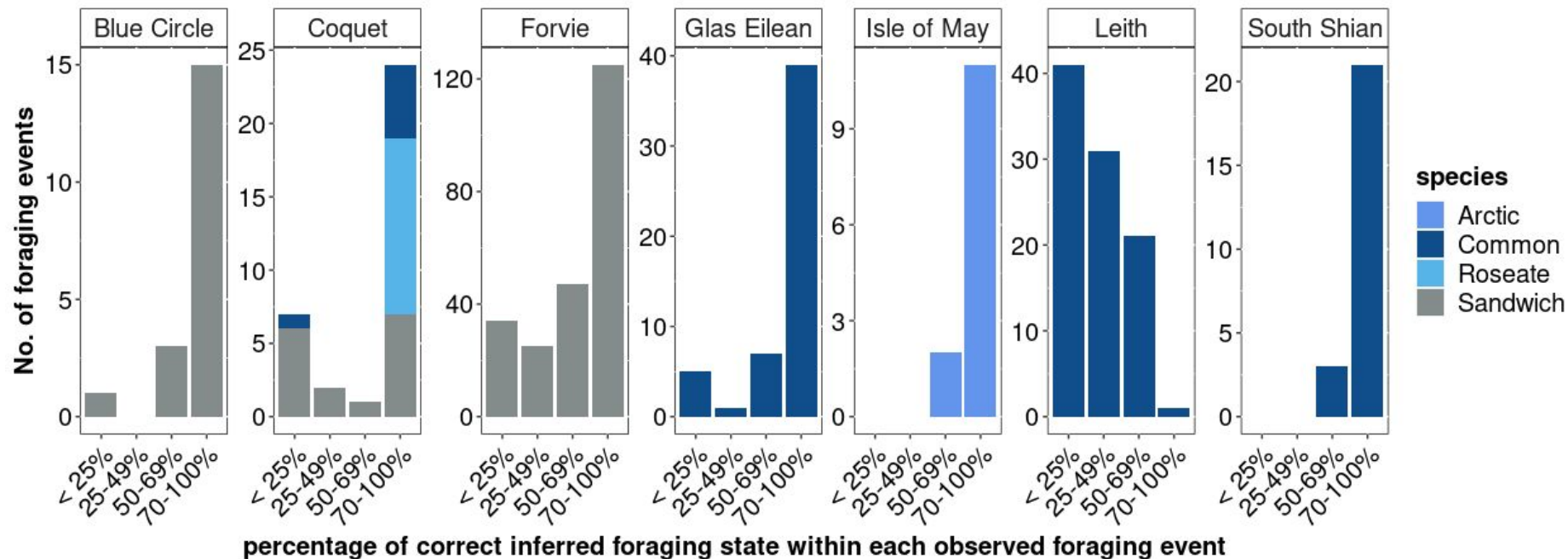


Figure 4: Proportion of correctly inferred foraging states within each observed foraging event across breeding colonies. Inferred foraging states are from HMMs deemed optimal based on validation.

Foraging events - incubation season

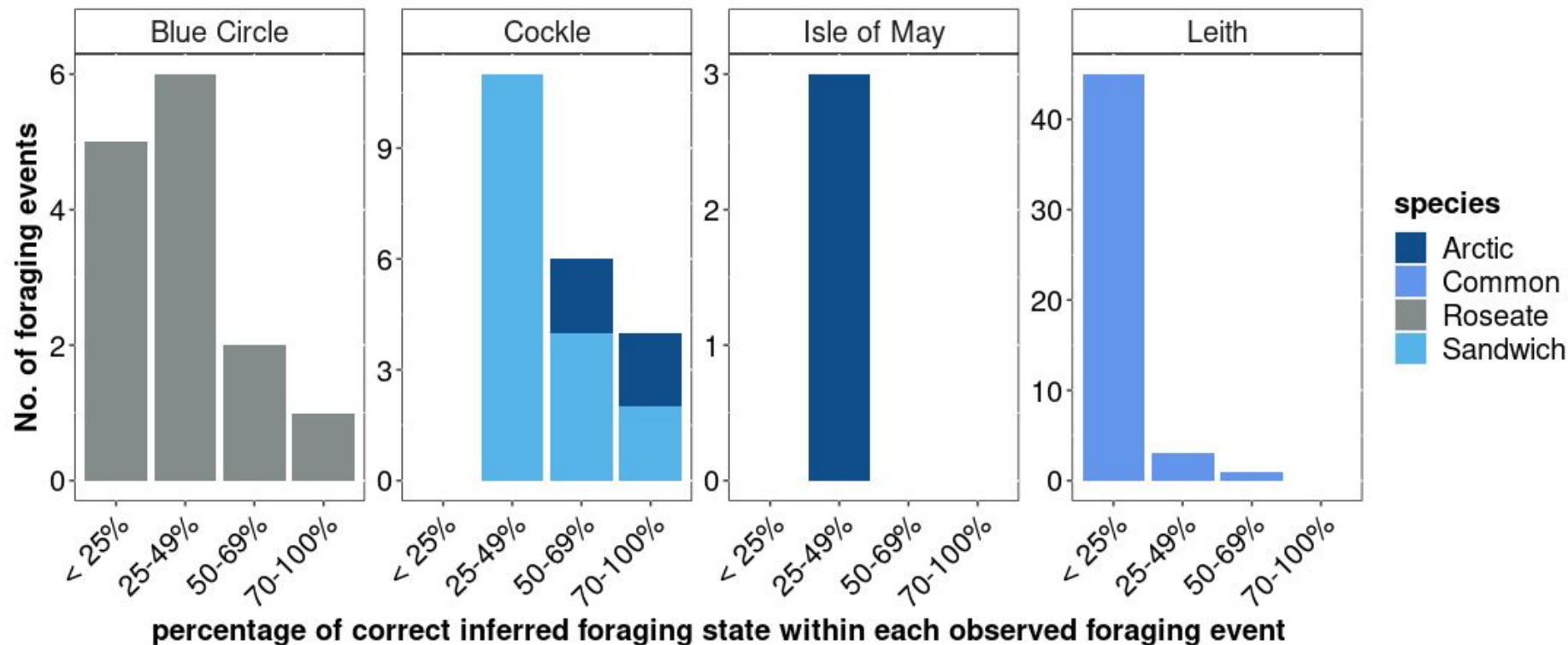


Figure 5: Proportion of correctly inferred foraging states within each observed foraging event across breeding colonies. Inferred foraging states are from HMMs deemed optimal based on validation.

Summary

- ❑ Visual tracking data is suitable for our study as boat tracks closely replicate those from the approximate tern tracks
- ❑ Important conservation-relevant behaviours inferred using HMMs is justified and can inform the design of designated protected areas
- ❑ There are no substantial differences in the performance of simpler and less simpler HMMs in inferring behavioural states

**THANKS
FOR
LISTENING**

QUESTIONS?